Ajinkya Jain

Senior Research Scientist, Google (Intrinsic) | Ph.D., UT Austin

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Research Profile

Senior Research Scientist at **Google (Intrinsic)** with **5+ years of post-PhD R&D experience** developing generalist manipulation policies, **on-policy distillation and RL finetuning**, **LLM/VLM policy synthesis**, and **force-aware imitation/online learning** for contact-rich tasks. Author of **11+ papers** across CoRL, ICRA, IROS, IEEE RA-L, and IEEE T-RL (**3,700+ citations**; **ICRA 2024 Best Conference Paper Award**), co-inventor of a granted US robotics patent, and research mentor to **9 PhD interns and graduate researchers**.

Education

The University of Texas at Austin — Ph.D., Mechanical Engineering (*Robotics & Controls*) 2015–2021
Advisors: Prof. Scott Niekum (Computer Science) & Prof. Ashish Deshpande

Indian Institute of Technology Kanpur — B.Tech.–M.Tech., Mechanical Engineering 2010–2015
Dual Degree in Robotics and Automation | Advisor: Prof. Bishakh Bhattacharya

Research & Industry Experience

Senior Research Scientist, Google (Intrinsic), Mountain View, CA Nov 2026–Present
Senior Software Engineer (Research), Google (Intrinsic), Mountain View, CA Oct 2023–Oct 2026
Robotics Engineer III (Research), Google (Intrinsic), Mountain View, CA Nov 2022–Sep 2023
R&D Team (Lead: Prof. Stefan Schaal)

- **RL Finetuning & Multi-Robot Policy Distillation:** Co-developed *RoboBallet2* (under review, *IEEE T-RL*), scaling collision-free multi-robot reaching across arbitrary 3D geometries via on-policy distillation and reinforcement finetuning in collaboration with Google DeepMind and UC San Diego.
- **Foundation Models & Compliant Policy Synthesis:** Co-designed *GenCHiP* (*IROS 2024*), equipping LLMs with compliant, force-constrained action spaces to synthesize policies for tight-tolerance, contact-rich assemblies.
- **Force-Aware Imitation & Online Adaptation:** Formulated online contact-force learning (*ICRA 2025*) and predictive force-conditioned imitation learning (*RSS 2026 Workshop*) for sub-mm connector insertion under uncertainty.
- **Multimodal & Cross-Embodiment Robot Learning:** Co-authored cross-lab research on sketch-conditioned manipulation policies (*RT-Sketch*, *CoRL 2024*), bimanual imitation learning benchmarks (*IEEE RA-L 2024*), and cross-embodiment datasets and RT-X models (*Open X-Embodiment*, *ICRA 2024 Best Conference Paper Award*).
- **Tech Transfer & Research Mentorship:** Architected generalizable manipulation, perception, and placement pipelines (**US Patent 12,330,304 B2**) across **10+ high-precision industrial assembly benchmarks**; scoped research agendas and mentored **6 research interns and 3 Master's students** (Stanford, CMU, UCSD, USC, ASU).

Robotics Engineer III, Vicarious FPC, Inc. (Acquired by Alphabet, Union City, CA) Oct 2021–Nov 2022

- Developed constrained manipulation, motion planning, and tight-fit insertion algorithms for robotic assembly and warehouse automation, minimizing cycle times while achieving $\geq 95\%$ task success.
- Devised data-driven 3D perception and in-hand pose estimation algorithms for high-precision object placement, establishing core IP transitioned into Google (Intrinsic) (**US Patent 12,330,304 B2**).

Graduate Research Assistant, Dept. of Computer Science, UT Austin (Austin, TX) Aug 2015–Aug 2021

- **Category-Independent Articulation Perception (*CoRL 2021, ICRA 2021*):** Invented *ScrewNet* and distributional depth-based screw-theory architectures to estimate 3D object articulation models directly from raw depth images without prior category labels (**2× data efficiency**).
- **Hybrid Kinematics Learning (*IROS 2020*):** Developed a changepoint-based method to learn planning-compatible hybrid kinematics of articulated objects from demonstrations (**up to 3× higher accuracy** vs. prior SOTA).
- **Manipulation Planning Under Uncertainty (*CoRL 2018*):** Formulated a trajectory-optimization POMDP planner leveraging contact interactions for long-horizon manipulation (**5× faster planning**, $\geq 90\%$ success).

Earlier Research Internships 2018–2019

- **AI Research Intern, Samsung AI Research Center – Robotics (NYC, Summer 2019):** Built *GrAB-Net*, an autonomous RGB-D imitation learning pipeline for category-level grasping across 40 unseen objects ($\sim 70\%$ success).
- **Robotics Research Intern, Vicarious FPC, Inc. (Union City, CA, Summer 2018):** Implemented task-agnostic visual servoing for high-precision (≤ 2 mm) assembly with fast convergence (≤ 20 s, $\geq 90\%$ success).

Selected Publications & Patents [Google Scholar: 3,700+ Citations]

Y. Zhi, M. Lai, **A. Jain**, G. Salhotra, K. Go, G. Ustensvik, T. Kroeger, S. Schaal, and M. C. Yip, "RoboBallet2: Multi-robot Reaching for Arbitrary Geometries via On-policy Distillation and Reinforcement Finetuning," *Under Review at IEEE Transactions on Robot Learning (T-RL)*, 2026.

Y. Korkmaz, K. Go, G. Salhotra, E. Biyik, **A. Jain**, and S. Schaal, "Predictive Force Conditioning for Contact-Rich Imitation Learning," *Robotics: Science and Systems (RSS) Workshop on RoboData*, 2026.

K. Tracy, Z. Manchester, **A. Jain**, K. Go, S. Schaal, T. Erez, and Y. Tassa, "Efficient Online Learning of Contact Force Models for Connector Insertion," *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

P. Sundaresan, Q. Vuong, J. Gu, P. Xu, T. Xiao, S. Kirmani, T. Yu, M. Stark, **A. Jain**, K. Hausman, D. Sadigh, J. Bohg, and S. Schaal, "RT-Sketch: Goal-Conditioned Imitation Learning from Hand-Drawn Sketches," *Conference on Robot Learning (CoRL)*, 2024.

K. Burns, **A. Jain**, K. Go, F. Xia, M. Stark, S. Schaal, and K. Hausman, "GenCHiP: Generating Robot Policy Code for High-Precision and Contact-Rich Manipulation Tasks," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

Open X-Embodiment Collaboration (incl. **A. Jain**), "Open X-Embodiment: Robotic Learning Datasets and RT-X Models," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. **[Best Conference Paper Award]**

M. Drolet, S. Stepputtis, S. Kailas, **A. Jain**, J. Peters, S. Schaal, and H. Ben Amor, "A Comparison of Imitation Learning Algorithms for Bimanual Manipulation," *IEEE Robotics and Automation Letters (RA-L)*, 2024.

A. Jain, S. Giguere, R. Lioutikov, and S. Niekum, "Distributional Depth-Based Estimation of Object Articulation Models," *Conference on Robot Learning (CoRL)*, 2021.

A. Jain, R. Lioutikov, C. Chuck, and S. Niekum, "ScrewNet: Category-Independent Articulation Model Estimation From Depth Images Using Screw Theory," *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

A. Jain and S. Niekum, "Learning Hybrid Object Kinematics for Efficient Hierarchical Planning Under Uncertainty," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.

A. Jain and S. Niekum, "Efficient Hierarchical Robot Motion Planning Under Uncertainty and Hybrid Dynamics," *Conference on Robot Learning (CoRL)*, 2018.

Granted Patent: A. Sriraman, **A. Jain**, and J. C. Wendelken, "Object Placement," *US Patent 12,330,304 B2*, Granted June 2025 (Google / Intrinsic).

Awards, Mentorship & Academic Service

- **Best Conference Paper Award**, IEEE International Conference on Robotics and Automation (ICRA), 2024 (*Open X-Embodiment*)
- **Research Mentorship — Google (Intrinsic):** Mentored 6 research interns and 3 Master's students (ASU academic collaboration) on robot learning, force adaptation, and multi-robot RL projects (2022–Present)
- **Second Runner-up Worldwide**, RoboCup@Home SSPL League (Team Austin Villa), Nagoya, Japan, 2017
- **Academic Service & Teaching:** Peer Reviewer for ICRA, IROS, CoRL, IEEE RA-L, and Mechatronics; Guest Lecturer & Graduate Teaching Assistant in Optimization, Controls, and Mechatronics (UT Austin & IIT Kanpur)

Technical Expertise

- **Robot Learning & Embodied AI:** Reinforcement Learning & RL Finetuning, On-Policy Distillation, Imitation Learning / Behavior Cloning, Foundation Models & LLMs/VLMs for Robotics, Policy Code Synthesis, Cross-Embodiment Datasets
- **Control, Planning & Perception:** Compliant Impedance & Force-Torque Control, Online Contact-Force Modeling, Trajectory Optimization (SNOPT, IPOPT), POMDP / Belief-Space Planning, Multi-Robot Coordination, 3D Depth & Pose Perception
- **Software & Hardware:** Python, C++, PyTorch, TensorFlow, MuJoCo, ROS, PyBullet, Gazebo, OMPL, MoveIt!, PCL, OpenCV, Kubernetes; 6/7-DOF Manipulators (UR, Kinova Gen3, Bimanual Setups, Toyota HSR, Force-Torque Sensors)